

Q: 2

Module 4 - Assignment

Machine Learning Assignment: Customer Segmentation & Purchase Prediction

Assignment Overview

You are a data analyst at an e-commerce company. The business team wants to understand customer behavior better to personalize marketing campaigns and predict future purchases. You have been provided with a customer transaction dataset containing customer demographics and purchase history.

Your task is to segment customers into meaningful groups and build a predictive model to forecast customer spending behavior.

Dataset Description

You will work with a **customer transaction dataset** containing the following columns:

- Customer Demographics:
- `customer_id`: Unique identifier for each customer
 - `age`: Age of the customer (years)
 - `gender`: Gender (Male/Female/Other)
 - `city_tier`: City classification (Tier 1, Tier 2, Tier 3)
 - `membership_type`: Type of membership (Silver/Gold/Platinum)
- Purchase Behavior:
- `total_spend`: Total amount spent by customer (in currency)
 - `num_transactions`: Number of transactions made
 - `avg_transaction_value`: Average value per transaction
 - `days_since_last_purchase`: Days since last purchase
 - `num_visits`: Number of website/app visits
 - `product_categories_purchased`: Number of different product categories purchased
 - `discount_used`: Total discount availed (in currency)
- Target Variable:
- `high_value_customer`: Binary label (1 = High value customer, 0 = Low value customer)
- [Download Dataset from here](#)

Part 1: Data Exploration and Preprocessing (20 points)

Tasks:

- 1.1 Load and Inspect the Data (5 points)
- Load the dataset using `pandas`
 - Display the first 10 rows
 - Check the shape of the dataset
 - Display data types and check for null values using `info()`
 - Generate summary statistics using `describe()`
- 1.2 Handle Missing Values (5 points)
- Identify columns with missing values
 - For numerical columns, fill with the median value
 - For categorical columns, fill with the mode
 - Verify that no missing values remain
- 1.3 Handle Duplicates (3 points)
- Check for duplicate rows based on `customer_id`
 - Remove duplicate rows if any exist
 - Report the number of duplicates removed (if any)
- 1.4 Basic Exploratory Analysis (7 points)
- Use `value_counts()` to show the distribution of:
 - `gender`
 - `membership_type`
 - `high_value_customer`
 - Create 2 visualizations using `seaborn` or `matplotlib`:
 - Histogram showing distribution of `total_spend`
 - Box plot comparing `total_spend` across different `membership_type`
 - Write 2-3 observations from your exploratory analysis

Part 2: Customer Segmentation using Clustering (25 points)

Tasks:

- 2.1 Prepare Data for Clustering (7 points)
- Select numerical features for customer segmentation:
 - `age`, `total_spend`, `num_transactions`, `avg_transaction_value`, `days_since_last_purchase`, `num_visits`
 - Apply Standard Scaling to normalize all selected features
 - Explain in 1-2 sentences why scaling is necessary before clustering
- 2.2 Determine Optimal Number of Clusters (8 points)
- Implement K-means clustering for K values ranging from 2 to 8
 - For each K value, calculate the Within-Cluster Sum of Squares (WCSS)
 - Create an elbow curve plot showing WCSS vs number of clusters
 - State your chosen K value and reasoning (2-3 sentences)
- 2.3 Apply K-means Clustering (10 points)
- Apply K-means clustering with your chosen optimal K value
 - Add the cluster labels as a new column `customer_segment` to your original dataframe
 - Display the count of customers in each cluster
 - Calculate and display the mean values of key features for each cluster using `groupby`
 - Write a brief profile (2-3 sentences) for each customer segment describing their characteristics

Part 3: Predictive Modeling (35 points)

Tasks:

- 3.1 Feature Engineering (5 points)
- Create 2 new features from existing data. Examples:
 - `spend_per_visit` = `total_spend` / `num_visits`
 - `discount_percentage` = `discount_used` / `total_spend`
 - Explain why these features might be useful (1-2 sentences)
- 3.2 Data Preparation for Modeling (10 points)
- Separate features (X) and target variable (y = `high_value_customer`)
 - Identify and apply One-Hot Encoding to categorical features (`gender`, `city_tier`, `membership_type`)
 - Split the data into training (80%) and testing (20%) sets with `random_state=42`
 - Apply Standard Scaling to numerical features:
 - Important: Fit the scaler only on training data to avoid data leakage
 - Transform both train and test sets
- 3.3 Build Classification Model (10 points)
- Build and train ONE classification model of your choice:
 - Option 1: Logistic Regression
 - Option 2: Decision Tree Classifier
 - Option 3: Random Forest Classifier
 - Fit the model on training data
 - Generate predictions on test data
 - Briefly justify your model choice (1-2 sentences)
- 3.4 Model Evaluation (10 points)
- Calculate and display:
 - Confusion Matrix
 - Accuracy
 - Precision
 - Recall
 - F1 Score
 - Create a classification report using `classification_report()`
 - Write 2-3 sentences interpreting the results (Which metric is most important for this business problem? How well is the model performing?)

Part 4: Model Optimization and Business Insights (20 points)

Tasks:

- 4.1 Hyperparameter Tuning (10 points)
- Define a parameter grid with 2-3 hyperparameters for your chosen model:
 - Decision Tree: `max_depth`, `min_samples_split`
 - Random Forest: `n_estimators`, `max_depth`
 - Logistic Regression: `C`, `penalty`
 - Use `GridSearchCV` with 3-fold cross-validation (`cv=3`)
 - Report the best parameters found
 - Evaluate the tuned model on test data
 - Compare performance (at least accuracy and F1-score) before and after tuning in a simple table
- 4.2 Feature Importance Analysis (5 points)
- Extract and display feature importance (for tree-based models) OR coefficients (for Logistic Regression)
 - Create a horizontal bar plot showing the top 5 most important features
 - Write 2-3 sentences interpreting what these features mean for the business
- 4.3 Business Recommendations (5 points)
- Provide 3 specific, actionable recommendations based on your analysis:
 - 1 recommendation from clustering insights
 - 1 recommendation from model predictions
 - 1 recommendation from feature importance
 - Ensure recommendations are data-driven and reference specific findings

Submission Guidelines

Submit one file:

Jupyter Notebook (.ipynb)

Filename: `FirstName_LastName_ML_Assignment.ipynb`

Must contain:

- All cells executed with visible outputs
- Markdown explanations for all written responses
- Labeled plots with titles and axis labels
- Clean, organized code with comments

Before Submitting:

- Restart kernel and run all cells from top to bottom
- Verify all outputs are visible
- Check that all markdown cells are properly formatted
- Import all libraries at the beginning

Mandatory Notebook Structure

```
# Machine Learning Assignment: Customer Segmentation & Purchase Prediction
Name:
Date:

## Part 1: Data Exploration and Preprocessing
### 1.1 Load and Inspect
### 1.2 Handle Missing Values
### 1.3 Handle Duplicates
### 1.4 Exploratory Analysis

## Part 2: Customer Segmentation
### 2.1 Prepare Data for Clustering
### 2.2 Determine Optimal k
### 2.3 Apply K-means and Profile Segments

## Part 3: Predictive Modeling
### 3.1 Feature Engineering
### 3.2 Data Preparation
### 3.3 Build Model
### 3.4 Model Evaluation

## Part 4: Optimization and Insights
### 4.1 Hyperparameter Tuning
### 4.2 Feature Importance
### 4.3 Business Recommendations
```

Code Quality Requirements

- Use meaningful variable names
- Include comments explaining key operations
- Ensure all cells run without errors in sequence
- Import all necessary libraries at the beginning
- Follow consistent formatting and style

Evaluation Criteria (Total: 100 points)

Component	Points
Part 1: Data Exploration and Preprocessing	20
Part 2: Customer Segmentation	25
Part 3: Predictive Modeling	35
Part 4: Optimization and Insights	20
Total	100

Note: Code quality and clarity of explanations are evaluated within each section.

Important Notes

- Plagiarism:** Any copied code will result in zero marks
- Partial Credit:** Available for partially completed sections with correct methodology
- Late Submission:** Follow the deadline strictly
- Questions:** Reach out during designated doubt-clearing sessions

Good luck!